Large Language Models (LLMs)

Large Language Models (LLMs) are advanced Artificial Intelligence models designed to **understand, generate, and manipulate human language**. They are trained on **very large datasets containing text from books, websites, articles, and other sources** so they can learn patterns in language and produce meaningful responses.

LLMs are a core technology in **modern AI systems**, powering applications like chatbots, virtual assistants, automated content generation, and coding assistants.

How LLMs Work

LLMs are typically built using **Transformer neural network architecture**. They process text by breaking it into smaller pieces called **tokens**.

Key working steps

1. **Tokenization**
   - Input text is split into tokens (words or parts of words).
   - Example:

     "Artificial Intelligence is powerful"

     Tokens:

     Artificial | Intelligence | is | powerful
2. **Training on Large Data**
   - LLMs are trained on billions or trillions of tokens.
   - The model learns **relationships between words and context**.
3. **Context Understanding**
   - The model analyzes the input and predicts the most probable next tokens.
4. **Text Generation**
   - The model generates responses based on probabilities learned during training.

OpenAI Models

Model	When to Use	Token Pricing (Approx per 1M tokens)	Summary
GPT-3.5	Simple chatbots, basic automation, small applications	Input: ~$0.50 Output: ~$1.50	Older model but still useful for low-cost applications and simple conversational tasks.
GPT-4	Complex reasoning, coding assistance, research tasks	Input: ~$30 Output: ~$60	High-quality responses with strong reasoning, but expensive compared to newer models.
GPT-4 Turbo	Production chatbots, enterprise tools, faster GPT-4 workloads	Input: ~$10 Output: ~$30	Faster and cheaper version of GPT-4 with similar capabilities.
GPT-4o	Real-time assistants, multimodal apps (text, image, audio)	Input: ~$2.50 Output: ~$10	Optimized for speed and multimodal AI applications.
GPT-4o Mini	High-volume chatbots, cost-efficient AI systems	Input: ~$0.15 Output: ~$0.60	Lightweight and very affordable model suitable for large-scale usage.
GPT-5	Advanced AI agents, deep reasoning, complex problem solving	Input: ~$2 Output: ~$15	Next-generation GPT model designed for improved reasoning and AI agent systems.
GPT-5.2	Enterprise AI systems, coding assistants, research tasks	Input: ~$1.75 Output: ~$14	One of the most capable OpenAI models with strong reasoning and efficiency.
o1	Logical reasoning, mathematics, scientific analysis	Input: ~$15 Output: ~$60	Specialized reasoning model optimized for step-by-step problem solving.
o1-mini	Affordable reasoning tasks, AI workflows	Input: ~$3 Output: ~$12	Smaller reasoning model offering lower cost with good reasoning performance.
o3	Advanced reasoning, AI agents, complex decision systems	Input: ~$2–$4 Output: ~$8–$16	Next-generation reasoning model designed for complex problem solving and multi-step reasoning.
GPT-5.2 Mini	High-volume chatbots, automation tools, cost-efficient AI apps	Input:~$0.50 Output: ~$2	Smaller and cheaper version of GPT-5.2 designed for scalable applications with moderate reasoning capability.
GPT-5.2 Nano	Lightweight AI applications, mobile apps, simple automation	Input:~$0.10 Output: ~$0.40	Very small and fast model optimized for low-cost AI workloads and simple tasks.

Top used models and when to use them:

- **GPT-4o**
  - Used for AI assistants, chatbots, automation tools
- **GPT-5 / GPT-5.2**
  - Used for advanced reasoning tasks
- **o1 / o3** : Good for mathematics, scientific analysis, and complex problem solving

Anthropic Claude Models

Model	When to Use	Token Pricing (Approx per 1M tokens)	Summary
Claude 1	Early enterprise chatbots, document summarization	Input:~$8 Output: ~$24	First Claude model focused on safe AI responses and enterprise use cases.
Claude 2	Research, document analysis, enterprise assistants	Input:~$8 Output: ~$24	Improved reasoning and longer context window than Claude 1.
Claude Instant	Fast chatbots, lightweight AI tasks	Input:~$0.80 Output: ~$2.40	Optimized for speed and cost efficiency compared to larger Claude models.
Claude 3 Haiku	High-volume chatbots, fast responses	Input:~$0.25 Output: ~$1.25	Smallest and fastest Claude 3 model designed for low-cost production workloads.
Claude 3 Sonnet	Enterprise applications, document processing	Input:~$3 Output: ~$15	Balanced model offering strong reasoning and long-context understanding.
Claude 3 Opus	Advanced reasoning, coding, research	Input:~$15 Output: ~$75	Most powerful Claude 3 model with strong reasoning and complex task handling.
Claude Sonnet 4	Enterprise AI systems, document analysis	Input:~$3 Output: ~$15	Updated Sonnet model optimized for enterprise workloads and long context tasks.
Claude Sonnet 4.6	Enterprise AI tools, document analysis, large-scale business workflows	Input:~$3 Output: ~$15	Updated Sonnet model with improved reasoning, better coding capability.
Claude Opus 4	High-end AI agents, research, complex reasoning	Input:~$5 Output: ~$25	Latest high-performance Claude model with improved reasoning and efficiency.

Top used models and when to use them:

- **Claude 3 Haiku**
  - Used for fast and cheap chatbots
  - Suitable for high-volume applications
- **Claude 3 Sonnet**
  - Used for enterprise AI tools
  - Good for document analysis and business workflows
- **Claude Opus**
  - Used for advanced reasoning and research

Google Gemini Models

Model	When to Use	Token Pricing (Approx per 1M tokens)	Summary
Gemini 1	EarlyGemini applications, chatbots, general AI tasks	Input:~$7 Output: ~$21	First generation Gemini model designed to replace earlier Google AI models like PaLM.
Gemini 1.5	Large document processing, research, long context tasks	Input:~$3.50 Output: ~$10.50	Improved version with significantly larger context window support.
Gemini 1.5 Pro	Complex reasoning, coding, large-scale document analysis	Input:~$3.50 Output: ~$10.50	High-performance model capable of processing extremely long inputs (up to ~1M tokens).
Gemini 2	Advanced AI assistants, multimodal applications	Input:~$2 Output: ~$8	Next generation Gemini model optimized for improved reasoning and efficiency.
Gemini 3	AI agents, advanced reasoning systems	Input:~$2 Output: ~$12	Newer generation model with stronger reasoning and better AI agent capabilities.
Gemini Flash	Real-time chatbots, fast responses, high-volume applications	Input:~$0.20 Output: ~$0.20	Lightweight model designed for speed and large-scale production workloads.
Gemini Flash Lite	Extremely cheap AI workloads, batch processing	Input:~$0.08 Output: ~$0.30	One of the lowest-cost LLM APIs suitable for large-scale data processing tasks.
Gemini Ultra	Advanced reasoning, enterprise AI systems, research applications	Input:~$10 Output: ~$30	Most powerful Gemini model designed for complex reasoning.

Top used models and when to use them:

- **Gemini Flash**
  - Used for cheap chatbots
  - Fast response times for large-scale applications
- **Gemini 1.5 Pro**
  - Used for analyzing long documents
  - Supports extremely large context windows
- **Gemini Ultra**
  - Used for complex reasoning and enterprise AI applications

xAI Grok Models

Model	When to Use	Token Pricing (Approx per 1M tokens)	Summary
Grok 1	Conversational AI, social media assistants	Input:~$5 Output: ~$15	First Grok model designed for conversational AI integrated with real-time internet data.
Grok 1.5	Improved chatbots, reasoning tasks	Input:~$4 Output: ~$12	Enhanced reasoning capabilities and faster responses compared to Grok 1.
Grok 2	AI assistants, analytics, enterprise AI applications	Input:~$2 Output: ~$10	More efficient and powerful model optimized for production AI systems.
Grok 2 Mini	High-volume chatbots, real-time assistants, cost-efficient AI applications	Input:~$0.50 Output: ~$2	Lightweight version of Grok 2 designed for faster responses and lower-cost AI workloads.
Grok 3	Advanced reasoning, AI agents, large-scale AI platforms	Input:~$1 Output: ~$8	Latest Grok model with improved reasoning, efficiency, and better performance for complex AI workflows.

Top used models and when to use them:

- **Grok 4** → Best for **advanced reasoning and complex AI systems**
- **Grok 3** → Good for **AI assistants and problem solving**
- **Grok 2** → Good for **general chatbots and coding support**

DeepSeek Models

Model	When to Use	Token Pricing (Approx per 1M tokens)	Summary
DeepSeek LLM	General AI tasks, chatbots, research projects	Input:~$0.27 Output: ~$1.10	Early DeepSeek language model designed for general conversational AI tasks.
DeepSeek V2	Enterprise AI applications, document analysis	Input:~$0.27 Output: ~$1.10	Improved version with better reasoning capability and efficiency.
DeepSeek V3	Large-scale AI systems, advanced reasoning tasks	Input:~$0.30 Output: ~$0.90	Highly optimized model offering strong performance at a very low cost.
DeepSeek Coder	Coding assistants, software development tools	Input:~$0.40 Output: ~$1.20	Specialized model designed for programming, code generation, and debugging tasks.

Top used models and when to use them:

- **DeepSeek V3**
  - Used for large-scale AI systems
  - Very cost-efficient model
- **DeepSeek Coder**
  - Used for programming tasks
  - Good for code generation and debugging
- **DeepSeek R1**
  - Used for mathematical reasoning

Industry Popular LLM Usage Table

Model	When to Use	Token Pricing (Approx per 1M tokens)	Summary
GPT-4 / GPT-5	AI assistants, coding tools, complex reasoning	Input:~$2.50 Output: ~$10	Powerful models used for advanced AI assistants, automation systems, and reasoning tasks.
Claude Sonnet	Enterprise AI tools, document analysis	Input:~$3 Output: ~$15	Balanced Claude model with strong performance in enterprise applications and long document processing.
Gemini Flash	Cheap chatbots, real-time AI apps	Input:~$0.20 Output: ~$0.20	Lightweight model optimized for speed and large-scale deployments at low cost.
LLaMA	Self-hosted AI systems, private enterprise AI	No token cost (self-host) Infrastructure cost only	Open-weight model that can run locally with full control over data privacy.
DeepSeek R1	Reasoning tasks, mathematics, scientific research	Input:~$0.27–$0.28 Output: ~$0.42	Advanced reasoning model optimized for logical problem solving and research tasks.
Mistral Large	Coding assistants, enterprise AI platforms	Input:~$8 Output: ~$24	High-performance model designed for coding and enterprise AI applications.
Mixtral	Open-source scalable AI systems	No API token cost (self-host)	Mixture-of-experts model providing strong performance with efficient compute usage.
Qwen 2.5	Multilingual enterprise AI, global applications	Input:~$0.40 Output: ~$1.20	Powerful multilingual model designed for enterprise AI deployments.
Phi-3	Edge devices, mobile AI assistants	No token cost (self-host)	Lightweight model optimized for small hardware and embedded AI systems.
Grok 3	Real-time analytics, conversational AI	Input:~$1 Output: ~$8 Pricing varies	Advanced Grok model optimized for large-scale AI applications and real-time responses.

In conclusion, there is no single “best” LLM for all scenarios. The most suitable model depends on the specific use case, including whether the goal is building AI assistants, enterprise tools, chatbots, reasoning systems, or private AI infrastructure. By carefully evaluating model capabilities, pricing, and deployment requirements, organizations can select the most appropriate LLM to build efficient and scalable AI applications.